

Project Synopsis:

To briefly summarize my goals and plans. I want to create a visualization of rain in Montgomery County overlaid with car crash information. My primary product will be a time-lapse showing a map of rainfall and car crashes. I may include other visualizations and inferences dependent on how long my primary objective takes to complete.

The Data:

In pursuit of my primary goal, I have two datasets. The one with which I will do the most work is a CSV (comma-separated values) file, which I requested from NOAA (National Oceanic and Atmospheric Administration). I have data for all of Maryland, but I have filtered only the information from stations in Montgomery County. The tool for requesting weather data from NOAA is linked [here](#)

My second dataset is from Data Montgomery and contains information on car crashes within Montgomery County. It is available through data Montgomery's portal [here](#)

Goals/Questions:

- How much and where did it rain in Montgomery County on a given day in 2025?
 - Method: I will create a map colored by rainfall to visualize rain patterns. I will use leaflet in R studio to create said map.
- How much did it rain at a given point that isn't one of my designated collection stations on a given day in 2025?
 - Method: I will use kriging to estimate the amount of rainfall in areas where I don't have exact data. To my understanding, kriging is a method of spatial interpolation. There is a R package called gstat designed for geospatial statistics which I can use to create what amounts to an image to overlay over my leaflet map.
- Where did Car crashes happen on a given day of 2025 in Montgomery County?
 - Method: This is fairly simple to visualize using leaflet in R.
- Is there a correlation between the location of heavy rainfall and location of car crashes on a given day in 2025 in Montgomery County?
 - Method: I can overlay the points from my map of car crashes onto my map of rainfall and then look for correlation.
- Is there a trend of correlation between the location of heavy rainfall and car crashes in Montgomery county over 2025?
 - Method: I can repeat the above processes for each day of 2025 then combining them to create a timelapse. I can then look for trends or patterns within the timelapse
- If a trend exists, why would such correlation exist?

- Method: It's pretty well known (at least if driving tests are to be believed) that rain can make the roads more slippery and hamper driver vision. Though in theory (If driving tests are being followed) you can slow down and increase following distance to mitigate the added danger of rain. So if I find clear correlation I might research into behavioral studies regarding following general advice regarding driving in rain.
- Which areas of Montgomery county have the most car crashes in 2025?
 - Method: I can make a map of all car crashes in 2025 and analyze the areas where car crashes are most and least likely.
- Why factors may cause this?
 - Method: Research, my main theory right now is regarding the difference between urban and rural communities and sheer amount of traffic on certain roads.

Risks and Mitigations:

The main risks I may run into are involved in the kriging process, I have never done any geospatial interpolation and thus I still need to continue reading to figure out all the little things I can tweak to make the most accurate model I can. There is also a mote of uncertainty in the method I am choosing to use I believe I have enough different collection stations' data to take a decently accurate picture but in the areas furthest away from stations there is no way to insure accuracy. Especially given the volatile nature of weather phenomena, which don't typically follow the smooth patterns that interpolation will likely give me. I can't really mitigate these problems but I can and will bring them up in my analysis as a sort of disclaimer.

Project Time Line:

March 24, 2026: Finish and finalize project outline

Marth 25-April 3, 2026: Continue reading and learning about kriging and the gstat package. Do some practice kriging and finalize parameters to keep consistent.

April 6, 2026: begin creating maps for specific days, applying kriging

April 6, 2026: begin compiling said maps to create timelapse

April 15, 2026: Create project report

April ?-May ?: finish compiling timelapse, research trends, depending on time create more visualizations or do more analysis.

May 6-12, 2026: finalize everything.

May 13, 2026: Submit report and give presentation.